Sizhe Wang

HomePage | sizhew@wustl.edu | Google Scholar

Education

Washington University in St. Louis — **08 / 2025 - Current**
PhD student in Computer Science (GPA: 3.93 / 4.0) — *St. Louis, U.S.*

University of Southern California — **01 / 2023 - 05 / 2025**
Master of Science in Computer Science (GPA: 3.86 / 4.0) — *Los Angeles, U.S.*

Sichuan University — **09 / 2018 – 06 / 2022**
Bachelor of Science in Physics — *Chengdu, China*

Research Experience

WashU | *LLM, AI for Health, Reinforcement Learning* — **10 / 2025 - Current**
Graduate Research Assistant, supervised by Prof. Chenyang Lu — *St. Louis, MO*

- Proposed **CURA** (Clinical Uncertainty Risk Alignment), an uncertainty calibration framework for clinical language models that aligns individual- and cohort-level risk estimates via a bi-level uncertainty fine-tuning scheme, thereby improving calibration of LM-based clinical risk prediction while maintaining competitive discriminative performance.
- Further showed that, with an uncertainty-based sampling strategy, the model can autonomously select only 5–20% of the most informative labeled cases for fine-tuning and still achieve performance comparable to full-dataset baselines, substantially improving calibration and data efficiency in clinical risk prediction tasks.

UNITES Lab, UNC | *LLM, RLHF, Alignment Tuning* — **09 / 2024 - 03 / 2025**
Research Assistant, supervised by Prof. Tianlong Chen — *Remote*

- Proposed Balanced Preference Optimization (BPO) to dynamically augment responses to diverse questions. BPO utilizes embedding-based clustering to sample representative questions and employs gradient-based clustering to select the most informative augmented samples based on the model's optimization direction. This framework, using only 10% of the original preference data, improved performance from 18.14% to 40.25% on HH-RLHF and from 36.84% to 48.24% on safeRLHF.
- Demonstrated that fine-tuning on only 1% of representative samples can lead to better performance than using the full dataset. Furthermore, applying a simple uniform balancing method yields additional improvements, increasing win rates on HH-RLHF from 18.14% to 25.34% and on safeRLHF from 36.84% to 46.18%.

UCSD | *LLM, Reasoning, Instruction Tuning* — **09 / 2023 - 05 / 2024**
Collaborative Student — *Remote*

- Developed a novel self-reinforcement methodology to enhance LLMs' reasoning capabilities with minimal human supervision. The approach iteratively improves models by learning from the differences between the responses from the SFT model and the base model on unlabeled questions using DPO. Validated the method's effectiveness through experiments on reasoning tasks.
- Implemented a new strategy, *learning from mistakes*, to enhance LLMs' reasoning capabilities by learning from errors rather than relying solely on accurate, handcrafted references. Demonstrated that this approach benefits both prompting and instruction tuning across various reasoning datasets.

Publications

CURA: Clinical Uncertainty Risk Alignment for Language Model–Based Risk Prediction (ACL 2026 Main)
Sizhe Wang, *Ziqi Xu, Claire Najjuuko1, Charles Alba, Chenyang Lu*

BPO: Towards Balanced Preference Optimization between Knowledge Breadth and Depth in Alignment (NAACL 2025 Main)
Sizhe Wang, *Yongqi Tong, Hengyuan Zhang, Dawei Li, Xin Zhang, Tianlong Chen*

Optimizing Language Model's Reasoning Abilities with Weak Supervision (arXiv preprint)
Yongqi Tong, **Sizhe Wang***, Dawei Li, Yifan Wang, Simeng Han, Zi Lin, Chengsong Huang, Jiaxin Huang, Jingbo Shang*
(*Equal contribution)

Can LLMs Learn from Previous Mistakes? Investigating LLMs' Errors to Boost for Reasoning (ACL 2024 Main)
*Yongqi Tong, Dawei Li, **Sizhe Wang**, Yujia Wang, Fei Teng, Jingbo Shang*

ShifCon: Enhancing Non-Dominant Language Capabilities with a Shift-based Contrastive Framework (ACL 2025 Main)

Hengyuan Zhang, Chenming Shang, ***Sizhe Wang****, Dongdong Zhang, Feng Yao, Renliang Sun, Yiyao Yu, Yujiu Yang, Furu Wei*

Eliminating Reasoning via Inferring with Planning: A New Framework to Guide LLMs' Non-linear Thinking (arXiv preprint)

Yongqi Tong, Yifan Wang, Dawei Li, ***Sizhe Wang****, Zi Lin, Simeng Han, Jingbo Shang*

Services

Reviewer: ACL Rolling Review (ARR); IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP); CIKM25

Technical Skills

Programming Languages: Python, Java, C, C++, JavaScript, Swift
Frameworks and Tools: PyTorch, vLLM, Verl, Hugging Face Transformers, SwiftUI, SQL, Google Cloud Platform (GCP)